

Repairing normal-anchor drift: three consolidation families

Stage 0: normal-only S-JEPA
(anchor c_0 frozen, label-free)

Stages 1-4: new pathology data
arrive one group at a time

Drift: running normal representation
moves away from c_0 (0.954 $\rightarrow$ 0.594)

representation space
Anchor distillation (AnchorGuard)
 $L = \lambda * (1 - \text{mean } \cos(z_n, c_0))$

implemented in notebook 08

parameter space
Anchor-directional EWC
penalize the top-k directions that move away

documented extension

data space
Weighted replay
weight normal batches (balanced replay already helps)

documented extension

Pre-registered success (notebook 08): stage-4 anchor ≥ 0.85 , feature std ≥ 0.35 (no collapse),
source-grouped probe macro-F1 within 0.05 of the canonical baseline - retention must not buy plasticity loss.